

BlueWave Island Tours Practice SBA

Full School-Based Assessment project brief

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: BlueWave Island Tours

Sector: Tourism

Description of Project

BlueWave Island Tours offers cultural, nature and adventure packages to visitors. The company wants to manage customer bookings, package details and payments, study booking trends and give customers clearer information before they arrive.

Build a tourism project that manages bookings, analyses revenue, produces customer documents, presents package information and automates final booking costs.

How to use this practice SBA

- Read the whole project before starting any software task.
- Complete the sections in order so information remains consistent from one file to the next.
- The wording and data are original SPARK practice material. Your teacher's current assignment takes priority.
- The practice task headings follow the style of common CSEC school assignments. Current CXC limits should still be followed.

Contents

Section	Practice weighting
Database Management	20 practice marks
Spreadsheet	20 practice marks
Word Processing	20 practice marks
Web Page Design	10 practice marks
Problem-Solving and Programming	15 practice marks

Database Management - 20 practice marks

Task A - Tables

1. Create a database named BlueWaveTours_YourFullName.accdb.
2. Create the Customer, TourPackage and Booking tables using suitable field names, data types and primary keys.
3. Enter realistic records and apply the validation rules required by the project.
4. Create the required relationships and enforce the correct links between the tables.

Task B - Queries and Form

1. Create a query called Upcoming Tours. Show bookings with TourDate on or after today. Include customer, package and tour date.
2. Create a query called Outstanding Payments. Join Customer, TourPackage and Booking and calculate BookingCost and Balance.
3. Create Customer Information Form with Booking subform and clear tour headings.

Task C - Report

1. Create Package Revenue Report grouped by PackageName with booking count and total BookingCost.

Spreadsheet - 20 practice marks

Task A - Calculations and Functions

1. Create TourBookings_YourFullName.xlsx with worksheets named Bookings and Dashboard. Enter or import the project data into Bookings and format it as a clear data table.
2. Create the main calculations using efficient formulas. The SPARK reference uses formulas such as $= (E2 * G2) + (F2 * H2)$, $= IF(E2 + F2 >= 5, I2 * 8\%, 0)$ and $= (I2 - J2) * (1 + \$M\$2)$.
3. Use at least three suitable functions. Functions used in the SPARK reference include VLOOKUP, COUNTIF, AVERAGE, IF, SUM.

Task B - Analysis and Presentation

1. Sort and filter the data to answer this requirement: Show bookings with Balance greater than 0 and TourDate within the next 30 days.
2. Create a summary showing Pivot-style summary of revenue by PackageName. Then create a suitable chart.
3. Create a linked total or key result on the Dashboard worksheet so that it updates when the detailed sheet changes.

Word Processing - 20 practice marks

Task A - Fillable Form

1. Create a document named Tour Booking Request Form_YourFullName.docx. The form must collect the information needed for this project and include clear instructions.
2. Use suitable form controls. The SPARK reference uses: Customer name text box; Tour date picker; Package drop-down list; Transport-needed check box.

Task B - Mail Merge

1. Create a professional main document called Booking Confirmation. Use the project data as the mail-merge source.
2. Insert the following merge fields where they make sense: CustomerName, PackageName, TourDate, BookingCost, Balance.
3. Complete the final merged output and include one relevant table, chart or other imported item from the connected SBA work where appropriate.

Web Page Design - 10 practice marks

Web Page Design

1. Create one web page for BlueWave Island Tours. The page must be suitable for the intended audience and include a project logo.
2. Organise the page into these content sections: Explore BlueWave, Tour Packages, What to Bring, Booking Information, Contact.
3. Add at least two suitable hyperlink types and test them.
4. Add at least one relevant graphic and make sure the page remains readable at different screen sizes.
5. Save, test and prepare the final web-page evidence.

Problem-Solving and Programming - 15 practice marks

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the final cost and balance for each tour booking and the combined value of bookings processed.
2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as TourBookings_YourFullName.pas.
2. Test the program and capture evidence that the expected and actual results agree.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

Submission checklist

- BlueWaveTours_YourFullName.accdb
- TourBookings_YourFullName.xlsx
- Tour Booking Request Form_YourFullName.docx
- Booking Confirmation_YourFullName.docx
- One completed web page with its supporting files
- TourBookings_YourFullName.pas
- ProgramDocumentation_YourFullName.pdf

Final reminder

Keep the same names, codes, dates, fees and rules across your database, spreadsheet, word-processing, web and programming work. A strong SBA reads like one connected project.